Cranial Nerves (V, VII, IX, X, XII): Set 2

1. The trigeminal nerve (CN V) is classified functionally as a mixed nerve. Which functional component is carried exclusively within its motor root to supply the muscles of mastication?
a. General Somatic Afferent (GSA)
b. General Visceral Efferent (GVE)
c. Special Visceral Efferent (SVE) / Pharyngeal Motor
d. Special Visceral Afferent (SVA)
e. General Visceral Afferent (GVA)
2. The mesencephalic nucleus of the trigeminal nerve is completely unique within the human central nervous system because it houses primary, unipolar sensory neurons inside the brainstem. It selectively mediates proprioception from the:
a. Skin of the forehead
b. Mucosa of the cheek
c. Muscles of mastication and periodontal ligaments
d. Anterior two-thirds of the tongue
e. Mucosa of the maxillary sinus
3. The facial nerve (CN VII) possesses a separate sensory and parasympathetic root known as the nervus intermedius (of Wrisberg). This specific root carries preganglionic parasympathetic (GVE) fibers destined for which peripheral ganglion?
a. Otic ganglion
b. Ciliary ganglion
c. Submandibular ganglion
d. Pterygopalatine and submandibular ganglia
e. Trigeminal ganglion
4. Taste sensations from the palate are mediated by special visceral afferent (SVA) fibers traveling through the greater petrosal nerve. The cell bodies of these primary sensory neurons are located in the:
a. Submandibular ganglion
b. Otic ganglion
c. Geniculate ganglion
d. Pterygopalatine ganglion
e. Superior ganglion of CN IX
5. The glossopharyngeal nerve (CN IX) exits the jugular foramen and gives off a tympanic branch (Jacobson's nerve). This branch enters the middle ear cavity to form the tympanic plexus, which eventually continues out of the temporal bone as the:
a. Greater petrosal nerve
b. Deep petrosal nerve
c. Lesser petrosal nerve
d. Chorda tympani
e. Auriculotemporal nerve
6. The lesser petrosal nerve carries preganglionic parasympathetic secretomotor fibers from the inferior salivatory nucleus. These fibers terminate by synapsing in the:
a. Submandibular ganglion
b. Pterygopalatine ganglion
c. Ciliary ganglion
d. Sympathetic chain

e. Otic ganglion

7. Postganglionic parasympathetic secretomotor fibers leaving the otic ganglion hitchhike along which nerve branch to regulate salivation from the parotid gland?

a. Long buccal nerve
b. Inferior alveolar nerve
c. Auriculotemporal nerve (branch of V3)
d. Chorda tympani
e. Temporal branch of CN VII

8. The superior salivatory nucleus in the pons gives rise to preganglionic parasympathetic fibers traveling within the facial nerve. Those fibers heading to the submandibular and sublingual glands branch off the main trunk via the:

a. Greater petrosal nerve
b. Lesser petrosal nerve
c. Deep temporal branch
d. Chorda tympani nerve
e. Posterior auricular nerve

9. The vagus nerve (CN X) carries general visceral efferent (GVE) parasympathetic fibers that control smooth muscle and glands throughout the thorax and abdomen. These preganglionic fibers originate in the:

a. Nucleus ambiguus
b. Solitary tract nucleus
c. Dorsal motor nucleus of the vagus
d. Salivatory nucleus
e. Edinger-Westphal nucleus

10. Skeletal muscles derived from the fourth and sixth pharyngeal arches (such as the pharyngeal constrictors and intrinsic laryngeal muscles) are supplied by the vagus nerve. The motor cell bodies for these SVE fibers sit in the:

a. Dorsal motor nucleus
b. Spinal nucleus of CN V
c. Hypoglossal nucleus
d. Nucleus ambiguus
e. Nucleus solitarius

11. All special visceral afferent (SVA) taste fibers carried by CN VII, CN IX, and CN X enter the medulla oblongata to terminate within the gustatory portion of the:

a. Nucleus ambiguus
b. Nucleus of the solitary tract (Nucleus solitarius)
c. Spinal nucleus of the trigeminal
d. Medial geniculate body
e. Inferior salivatory center

12. The hypoglossal nerve (CN XII) consists purely of General Somatic Efferent (GSE) motor fibers. As it exits its cranial canal, it runs downward and forward, passing superficial to the internal and external carotid arteries, deep to the:

a. Posterior belly of the digastric and stylohyoid muscles
b. Mylohyoid muscle sheets completely
c. Investing layer of deep cervical fascia
d. Anterior belly of the digastric
e. Sternocleidomastoid muscle core

13. During its course in the neck, the hypoglossal nerve is joined briefly by a loop of nerve fibers that do not originate from the brainstem. These hitchhiking fibers come from the:
a. Sympathetic trunk
b. Facial nerve trunk
c. C1 spinal nerve ventral ramus
d. Brachial plexus trunks
e. Glossopharyngeal nerve root
14. Fibers derived from the C1 spinal segment hitchhike along the hypoglossal nerve and leave it as separate branches to supply two specific neck muscles. These are the:
a. Sternohyoid and sternothyroid
b. Thyrohyoid and geniohyoid muscles
c. Mylohyoid and anterior digastric
d. Stylohyoid and posterior digastric
e. Omohyoid superior and inferior bellies
15. When a patient presents with an upper motor neuron (UMN) lesion of the facial nerve (such as from a stroke in the motor cortex), which clinical manifestation is seen on the face?
a. Total paralysis of the entire ipsilateral side of the face
b. Total paralysis of the entire contralateral side of the face
c. Paralysis of only the lower quadrant of the face on the contralateral side, with sparing of the forehead
d. Sparing of the lips, with complete loss of eye closure bilaterally
e. Complete loss of taste, with normal muscle function everywhere
16. Sparing of the upper facial muscles (forehead wrinkling and eye closure) in an UMN facial nerve lesion occurs because the motor neurons in the facial nucleus supplying the upper face receive:
a. No cortical representation at all
b. Bilateral corticobulbar innervation from both cerebral hemispheres
c. Direct supply from the spinal cord segments
d. Only ipsilateral cortical tracks
e. Motor control via the trigeminal nerve
17. The mandibular nerve (V3) gives off the lingual nerve, which runs through the infratemporal fossa. Before it enters the oral cavity, the lingual nerve is joined by the chorda tympani, which introduces:
a. GSA fibers for pain
b. SVA (taste) and GVE (parasympathetic) fibers from CN VII
c. SVE motor fibers for mastication
d. Sympathetic fibers from the internal carotid plexus
e. GSE fibers for tongue movement
18. The glossopharyngeal nerve (CN IX) provides general somatic sensation (pain, touch, temperature) to the mucosa of the palatine tonsils and oropharynx. This pathway forms the afferent limb of the:
a. Corneal reflex
b. Gag reflex
c. Pupillary reflex
d. Jaw-jerk reflex
e. Lacrimal reflex

19. The efferent motor limb of the gag reflex, which drives the rapid, protective contraction of the pharyngeal constrictor muscles, is mediated by SVE fibers of the:
a. Trigeminal nerve (V3)
b. Facial nerve (CN VII)
c. Glossopharyngeal nerve (CN IX)
d. Vagus nerve (CN X via the pharyngeal plexus)
e. Hypoglossal nerve (CN XII)
20. The tensor tympani muscle inside the middle ear acts to dampen loud sound vibrations on the tympanic membrane. This muscle receives its motor innervation from the:
a. Facial nerve (CN VII)
b. Mandibular nerve (V3 branch of Trigeminal)
c. Glossopharyngeal nerve (CN IX)
d. Chorda tympani nerve
e. Vagus nerve (CN X)
21. The stapedius muscle works inside the middle ear to dampen excessive movements of the stapes footplate. This muscle is supplied by a branch arising from the:
a. Mandibular nerve (V3)
b. Maxillary nerve (V2)
c. Facial nerve (CN VII)
d. Vestibulocochlear nerve (CN VIII)
e. Glossopharyngeal nerve (CN IX)
22. Let's do a brief touch on the functional components of other nerves. Cranial Nerve I (Olfactory) and Cranial Nerve II (Optic) carry special sensory signals. They are classified structurally as:
a. General Somatic Efferent channels
b. Special Visceral Efferent channels
c. General Visceral Afferent channels
d. Special Somatic Afferent or Special Visceral Afferent channels
e. Pharyngeal motor tracks
23. Cranial Nerve III (Oculomotor) contains preganglionic parasympathetic fibers that control pupillary constriction and accommodation. These fibers originate inside the midbrain within the:
a. Nucleus ambiguus
b. Inferior salivatory nucleus
c. Edinger-Westphal nucleus
d. Superior salivatory nucleus
e. Dorsal motor nucleus
24. Cranial Nerve IV (Trochlear) is unique because it is the only cranial nerve that emerges from the dorsal aspect of the brainstem. It supplies which single extraocular muscle?
a. Lateral rectus
b. Superior oblique muscle
c. Inferior oblique muscle
d. Superior rectus
e. Medial rectus

25. Cranial Nerve VI (Abducens) has a long, vulnerable intracranial course over the clivus. It passes directly through the center of the cavernous sinus to supply the:
a. Medial rectus muscle
b. Superior oblique muscle
c. Lateral rectus muscle
d. Orbicularis oculi
e. Levator palpebrae superioris
26. Cranial Nerve XI (Spinal Accessory) takes origin from motor neurons located within the ventral horn of the upper five or six cervical spinal cord segments. These fibers ascend through the foramen magnum and exit the skull via the:
a. Foramen ovale
b. Jugular foramen
c. Hypoglossal canal
d. Foramen rotundum
e. Stylomastoid foramen
27. General somatic sensation (pain and touch) from the skin of the external acoustic meatus and tympanic membrane is shared by branches of the trigeminal nerve and the auricular branch of the:
a. Facial nerve
b. Vagus nerve (Arnold's nerve)
c. Glossopharyngeal nerve
d. Hypoglossal nerve
e. Accessory nerve
28. Stimulation of the auricular branch of the vagus nerve (Arnold's nerve) by a speculum or cold water inside the external ear canal can occasionally trigger reflex coughing, vomiting, or fainting due to:
a. Primary activation of the tongue muscles
b. Reflex vagal bradycardia and parasympathetic activation
c. Sudden high-pressure arterial flows
d. Intermittent spasms of the masseter muscle
e. Direct stimulation of the olfactory bulb
29. The internal laryngeal nerve is a direct branch of the superior laryngeal nerve from CN X. It pierces the thyrohyoid membrane to carry sensory information from the larynx mucosa:
a. Inferior to the vocal folds
b. Superior to the vocal folds (mediating the cough reflex)
c. From the trachea rows exclusively
d. From the pharyngeal tonsils base
e. Directly inside the thyroid gland stroma
30. The recurrent laryngeal nerve supplies all intrinsic muscles of the larynx except one. This single muscle, which tenses the vocal folds and is supplied by the external laryngeal nerve, is the:
a. Posterior cricoarytenoid
b. Cricothyroid muscle
c. Lateral cricoarytenoid
d. Thyroarytenoid
e. Aryepiglottic muscle